Kunming Shao

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EDUCATION

B.S. Purdue University, West Lafayette, IN **2017 - 2021**
Major: Biological Engineering / Cellular and Biomolecular Engineering
Minor: Biotechnology

Ph.D Candidate Purdue University, West Lafayette, IN **2022 - Present**
Purdue University Interdisciplinary Life Science Program (**PULSe**)
Department of Biological Sciences
Major - Computational and Systems Biology

PUBLICATIONS

Chang Y***, Shao K***, Li H, Jin G, Bentley RT, McCain RR, Crain CJ, Shen J, Tan Y, Liang PY, Harper HA, Torregrosa-Allen S, Elzey BD, Vanhaezebrouck IF, Cohen-Gadol AA, Dong C, Zhu Y, Wang Y, Luo J, Lian XL, Bao X. CAR-neutrophils produced in vivo to treat glioma. Nature Biomedical Engineering. 2026. doi:10.1038/s41551-026-01656-0 (*Co-first authors)

Luong NH*, **Shao K***, Duong VT, Bao X, Lin CC. Photo-click Decellularized Matrix Hydrogels for Generating Pancreatic Ductal Organoids. bioRxiv [Preprint]. 2026 Feb 20:2026.02.16.706185. doi: 10.64898/2026.02.16.706185. PMID: 41757071; PMCID: PMC12934964. (*Co-first authors)

Chiu HH*, Varghese A*, **Shao K***, Lu YC, Nahar R, Chen H, Deng Q, Bao X, Li C. scChat: A large language model-powered co-pilot for contextualized single-cell RNA sequencing analysis. AIChE J. 2026 Feb 11:e70285. doi:10.1002/aic.70285. (*Co-first authors)

Wang T, Kim DH, Ding C, Wang D, Zhang W, Silic M, Cheng X, **Shao K**, Ku T, Zheng C, Xie J, Xiao S, Jayant K, Yuan C, Chubykin AA, Staiger CJ, Zhang G, Deng Q. Inwardly rectifying potassium channels promote directional sensing during neutrophil chemotaxis. J Cell Biol. 2026 Jan 5;225(1):e202503037. doi: 10.1083/jcb.202503037. Epub 2025 Nov 19. PMID: 41259080; PMCID: PMC12629209.

Yan B, Wang C, Chakravorty S, Zhang Z, Kadadi SD, Zhuang Y, Sirit I, Hu Y, Jung M, Sahoo SS, Wang L, **Shao K**, Anderson NL, Trujillo-Ochoa JL, Briggs SD, Liu X, Olson MR, Afzali B, Zhao B, Kazemian M. A comprehensive single cell data analysis of lymphoblastoid cells reveals the role of super-enhancers in maintaining EBV latency. J Med Virol. 2023 Jan;95(1):e28362. doi: 10.1002/jmv.28362. PMID: 36453088; PMCID: PMC10027397.

Jin G, Fang S, Chang Y, **Shao K***, Li H, Tan Y, Lian X, Deng Q, Huang Y, Bao X. Targeting Cardiac Fibrosis with Chimeric Antigen Receptor Neutrophils Derived from Human Pluripotent Stem Cells. bioRxiv. 2026 Oct 13. doi:10.64898/2026.10.11.758383.

Shao K*, Wang M, Zheng C, Bao X, Deng Q. ANTS: a 3D-printed end-to-end platform for automated quantification of neutrophil-like cell migration. Manuscript submitted, 2026.

Shao K*, Liang Z, Shen J, Jin G, Li H, Tan Y, Zheng C, Lian X, Deng Q, B X. Chemically defined production of CAR-neutrophils from human pluripotent stem cells. Nature protocol, under review, 2026.

Ding C*, **Shao K***,Keren C, Li L, Zheng C, Adrian B, Zhang C, Umlis D, Deng Q. Piezo1 orchestrates zebrafish epithelial wound repair through mechanosensitive signaling. Manuscript submitted, 2026.

Shao K*, B X, Deng Q. How Mature Is Mature Enough for iPSC-Derived Neutrophils? Manuscript submitted, 2026.

RESEARCH EXPERIENCE

- **Graduate Research Assistant.**
  Research Associate, Advisor: Dr. Qing Deng, Department of Biological Science
  April 2023 – Present
  - Elucidated microRNA-mediated regulatory networks governing the development and effector functions of iPSC-derived neutrophils
  - Refined and scale up the iPSC-to-neutrophil differentiation pipeline, bone marrow organoid, to enhance cell yield, purity, and functional maturity

  Research Associate, Co-Advisor: Dr. Xiaoping Bao, Department of Chemical Engineering
  April 2023 – Present
  - Engineered neutrophil-like cells via nanoparticle-mediated delivery of modRNA
  - Integrated machine learning pipelines to guide the rational design and optimization of CAR-engineered iPSC-neutrophils; CAR-NK; iPSC-NK.
  - Conduct in vivo and ex vivo studies to characterize engineered immune-cell phenotype, trafficking, and antitumor activity.

- **Undergraduate Research Assistant.**
  Research Associate, Advisor: Dr. Majid Kazemian, Department of Biochemistry
  August 2019 – Dec 2021
  - Working on the novel noncoding RNA in viral associated cancers

- **Undergraduate Research Assistant.**
  Research Associate, Advisor: Dr. Luo Zhaoqing, Department of Biological Sciences

  Dec 2018 – Dec 2019
  - Study the mechanisms of how *L. pneumophila* modulates apoptotic pathways to promote its intracellular growth.

HONORS AND AWARDS

- Recipient of Presidential Scholar - Graduate Student 2026
- Recipient of the OIGP Bilsland Dissertation Fellowship 2026-2027
- Recipient of the Lynn Fellowship 2022
- Recipient of the 2021-2022 Ladisch ABE Scholarship 2021
- Recipient of the 2020-2021 Alumni & Friends of ABE Scholarship 2020
- Purdue Soybean Innovation Competition Award 2020
- Dean's List and Semester Honors, College of Engineering 2018 – 2021

SKILLS

- **Programming:** Python, R, MATLAB, SAS, SQL, Linux/Unix Shell Scripting (Bash), LaTeX, ImageJ Macro, VBA, Git

- **Software:** FlowJo, FIJI/ImageJ, IGV, JMP, Jupyter Notebook, Google Colab, GraphPad Prism, Seurat, Scanpy, Cell Ranger, DESeq2, QuPath, Cellpose, SnapGene, DNA Master

- **Molecular and Cellular Techniques:** PCR, qPCR, RT-qPCR, DNA/RNA extraction, molecular cloning, immunofluorescence (IF), immunohistochemistry (IHC), confocal microscopy, proximity labeling (TurboID/BioID-based biotinylation and streptavidin enrichment), ELISA, gel electrophoresis, mammalian cell culture, CRISPR/Cas9-mediated mammalian cell gene knock-in and out, Nucleic Acid Engineering (modRNA/gRNA design, IVT synthesis, and structural/integrity QC), LNP/Nanoparticle Formulation, western blotting(WB), flow cytometry, FACS sorting, nanoparticle formulation and characterization, extracellular vesicle (EV)/exosome isolation and characterization

- **Stem Cells:** Human and mouse hPSC/iPSC culture and directed differentiation; ipsc derived NK cells; ipsc derived dendritic cells; ipsc derived neutrophils; organoid establishment and 3D culture; CRISPR/Cas9 genome engineering with targeted knock-in/knockout at the AAVS1 safe-harbor locus; AAV and lentiviral vector design, and stable cell-line generation; primary T cell isolation, activation, expansion, and culture; CAR-T,CAR-NK and CAR-neutrophil end to end manufacturing; and functional characterization.

- **Omics:** bulk RNA-seq, ATAC-seq, ChIP-seq, single-cell RNA-seq (scRNA-seq) , Sample preparation, data processing and analysis

- **In vivo Animal Model:** *Mus musculus*; *Danio rerio*

- **In Vivo & Disease Models:** Glioma, and prostate cancer xenograft models; acute lung inflammation, bacterial pneumonia, and pulmonary fibrosis models; neutrophil trafficking and recruitment; immune-cell administration; tissue harvesting and processing; histology, imaging, CFU quantification, and efficacy assessment.

- **Artificial Intelligence & Deep Learning:** PyTorch, LLM application engineering (API integration, Agentic engineering, RAG workflows, and local model deployment for biological data analysis)